Amaan Khan Contact: 0305-4716616	**BT 404 Current Final Terms Spring 2020**

2 marks

Qno1: Define Antioxidants.

Antioxidants are food additives used to stabilize foods against oxidation. Oxidative quality changes in foods include the development of rancidity from the oxidation of unsaturated fats resulting in off-odors and off flavors and discoloration from oxidation of pigments or other components of the food.

Qno2: Define HACCP.

HACCP is a systematic preventive approach to food safety from biological, chemical and physical hazards in production processes that can cause finished product to be unsafe and designs measures to reduce these risks to a safe level.

Qno3: Write down the importance of Vitamin A.

Carotene conents in plant foods convert to Vitamin A. Important for vision and resistance to infection. It is also play important role in bone and teeth formation.

Qno4: Write down names of 4 factors affecting Microbial growth.

Factor affecting Microbial growth are

◊ pH

◊ Moisture

◊ Humidity of environment

◊ Temperature

◊ Availability of Oxygen

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3 marks

Qno1: Under what conditions fruit and vegetables must be stored?

Fruits and vegetables must be stored under

◊ Low oxygen

◊ High CO_2

◊ Proper storage temperature

Qno2: Why Sulphur dioxide is used in food processing?

Sulfur dioxide is used in food processing because of reasons given below;

• Used to prevent browning in certain dried fruits

• Prevent wild yeast growth in wines used to make vinegar

Qno3: What are acidulants name?

• Citric Acid - Citrus family

• Malic Acid - Apples

• Acetic Acid – Vinegar

Qno4: Write down the unique properties of tertroxidant?

I couldn't find answer of this question, because I think it is wrong.

Qno5: Give some examples of microorganisms.

Bacillus subtilis

Staphylococcus aureus

Lactose reuteri

5 marks

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Qno1: Write a note GMP practices.

(GMPs) that have been mandated by government agencies on the basis of scientific knowledge relating to known health hazards in foods, and the need to prevent unacceptable levels of these hazards or to eliminate them from food. GMP programs or they should cover the basic controls required for raw materials, ingredients, packaging materials and products, and for the plant's facilities, employees, equipment, operations, and environment that influence the safety of a food.

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Some GMPs Addresses

◊ Quality assurance consequently

◊ Traditional GMPs, along with some additional requirements, are now universally regarded as Prerequisite Programs for the implementation of an HACCP system.

Components of GMP

1) GHP (Good Hygiene Practices

2) GPP (Good Personnel Practices

3) GVP (Good Veterinary Practices Healthy, Fit, Clean animal

4) Good Sanitary Practices

5) Good raw material practices

◊ No cross contamination

◊ Proper storage

◊ Air, humidity ventilation

6) Good environment practices

Qno2: Write down a note on flavoring and flavor enhancers.

Flavorings:

Flavorings can be natural or synthetic, that are added to foods to produce flavors.

Flavor Enhancers:

Intensify existing flavors. Monosodium glutamate (MSG) is one of the best known and most widely used flavor enhancers.

Qno3: Describe the importance of water in food.

Importance of Water in Foods

1. Water as Quality Factor in Preservation:

Such as in products like dried milk, dehydrated vegetables, spices and herbs.

2. Quality factor of products:

Such as jams jellies, fruit syrups, prepared cereals etc.

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3. Convenience in Transportation:

Reduced moisture food is helpful in packaging and shipping such as concentrated milk, liquid cane sugar, and concentrated fruits juices.

4. Moisture as compositional standards Such as in foods cheese, flour, pineapple, meat etc.

5. Moisture as nutritional Standards

6. Moisture data as expressed in analytical results of many foods such as food contents on wet basis and dry basis

Water Activity is a concept different than water contents and it has important implications such as non-enzymic browning, lipid oxidization, degradation of vitamins, enzymatic hydrolysis, protein denaturation, and starch gelatinization and retrogradation. Shelf life, Microbial growth.

Qno4: Explain Oxidizing enzyme in food biotechnology.

• Loss of electrons in an atom or simply a chemical combination with oxygen.

• Such as wick of a candle burning.

• Oxidizing enzymes that bring about changes in foods that result in deterioration due to oxidation.

• In plants, peroxidases, ascorbic acid oxidase, tyrosinase, and polyphenolases may cause undesirable chemical reactions to occur.

• Peroxidases may oxidize certain phenol like compounds in root vegetables, such as horseradish, causing the prepared product to become darker in color when damaged.

• Ascorbic acid oxidase, in vegetables, oxidizes ascorbic acid cause loss of the vitamin C content of foods.

• Peroxidases aIso, indirectly, cause a loss of vitamin C by its by products.

• Phenolases are present in some fruits and vegetables cause brown or dark-colored compounds to be formed when the tissues are cut.

• The enzyme tyrosinase, cause discoloration of the cut tissue and will also oxidize compounds related to tyrosine.

• This enzyme is also present in shrimp and some spiny lobsters and may cause a discoloration called black spot as a black stripe on the flesh along the edges of the segments of the tail or as a pronounced band where the shell segments overlap.

Qno5: Write a note on transgenic plants.

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Involves the process of incorporating the selected desired genes into plants and then enabling these altered plants to produce the encoded proteins The process is known as transformation Altered plants are known as transgenic plants.

Transgenic potatoes for diarrhea:

• First successful human trial was conducted in 1997

• Transgenic potatoes, which possessed the subunit of the E. coli heat-labile toxin, responsible for diarrhea

• 4-fold increase in serum antibodies was manifested in ten out of the 11 volunteers

Transgenic tomatoes against diarrhea:

• Transgenic tomatoes were produced against the Norwalk virus

• Responsible agent for severe diarrhea.

• The transgenic tomatoes are capable to produce surface protein specific to the virus.

• Mice fed with transgenic tomatoes showed immune response towards the virus.

Other transgenic plants:

Presently, banana is being exploited as a good source for edible vaccine production

• It does not require cooking

• Locally grown plant

10 marks

Qno1: What changes take place after animal slaughter?

When the animal is slaughtered certain type of changes starts to take place:

• Stiffening of meat called Rigor mortis

• Formation of lactic acid lowering pH

• Swelling of muscle fibers with fluid causing the meat to become hard.

When the animal is slaughtered certain type of changes starts to take place:

• The circulatory system stops functioning but the enzymes continue to remain active.

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• Oxidation of glycogen brings about production of lactic acid.

• The amount of glycogen stored in a particular animal decides the amount of lactic acid produced after slaughter.

• A lower glycogen storage produces sticky and gummy meat which is undesirable.

• Stiff muscles of the animal start softening after holding for some time. This process is called aging.

• Tenderness in the meat increases with holding time. Holding conditions should be carefully controlled.

Qno2: Explain the types of botulism.

There are five types of botulism:

1. Food-borne botulism

2. Wound botulism

3. Infant botulism

4. Adult infectious botulism

5. Inadvertent botulism

♣ Salmonella is a Gram negative, rod shaped,

♣ Motile bacterium that does not form spores. It is found in fresh and

♣ Salt water, soil and animal faeces. It causes salmonellosis

♣ Infection and even extremely low doses can cause death.

Listeria monocytogens are Gram-positive, motile bacteria. They do not form spores, but are quite resistant to desiccation and heat. Found in the soil and associated with birds and animals Cause disease called listeriosis, which can result in septicaemia, meningitis encephalitis and spontaneous abortions in pregnant females.

S. aureus is a Grampositive, small round bacterium (coccus) that can produce a heat-stable toxin. S. aureus is found all over most commonly via humans. It is found in the nasal passages, throat, on the skin and in the hair of more than half of healthy people.

Escherichia coli are Gram-negative bacteria found in the intestines of all mammals. It cause four different classes of illnesses. Outbreaks are associated with fecal contamination of water by food handlers, or are due to unprocessed or under-processed foods.

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Campylobacter jejuni are Gram-negative, microaerophilic, motile, rod- shaped bacteria. They are found on raw chickens, in raw milk, and are often carried in the intestines of healthy cattle. They can contaminate food via flies on farms and are also found in non chlorinated water.

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